DC Pass

Power Splitter/Combiner ZX10-2-1252-S+

2 Way-0° 50Ω 1800 to 12500 MHz

The Big Deal

- Ultra-Wideband, 1800 to 12500 MHz
- Low insertion loss, 0.7 dB
- Low amplitude unbalance, 0.1 dB
- Rugged unibody case

CASE STYLE: FL2227

Product Overview

Mini-Circuits' ZX10-2-1252-S+ is a coaxial, ultra-wideband 2-way 0˚ splitter/combiner providing RF input power handling up to 1.85W as a splitter and 0.7 dB insertion loss for an extremely wide range of applications from 1800 to 12500 MHz. Its outstanding combination of low loss and low unbalance make this model an excellent choice for distributing signals in systems where efficient transmission of signal power is needed. The splitter/combiner comes housed in a rugged, compact case (0.74 x 0.90 x 0.54") with SMA connectors.

Key Features

Feature	Advantages
Ultra-wideband, 1800 to 12500 MHz	ZX10-2-1252-S+ supports bandwidth requirements for a wide variety of applications including broadband applications such as instrumentation and defense.
Low insertion loss, 0.7 dB	Provides excellent transmission of signal power, making this model an excellent candidate for signal distribution applications where low loss is a requirement.
Low amplitude unbalance, 0.1 dB	Produces nearly equal output signals, ideal for parallel path / multichannel systems.
DC passing up to 400mA	Supports applications where DC power is needed through the RF line.
Rugged, unibody construction	Mini-Circuits' unibody construction integrates the RF connector into the case body, providing high reliability and excellent survivability in critical applications.

Notes
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B. Electrical specifications and performance data contained in this specification document are based on Mini-Circuit's applicable established test performance criteria and measurement instructions.
C. The parts covered by this specification document are subject to Mini-Circuits standard limited warranty and terms and conditions (collectively, "Standard Terms"); Purchasers of this part are entitled to the rights and benefits contained therein. For a full statement of the Standard Terms and the exclusive rights and remedies thereunder, please visit Mini-Circuits' website at www.minicircuits.com/MCLStore/terms.jsp

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Generic photo used for illustration purposes only

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Connectors	Model
SMA	ZX10-2-1252-S+

+RoHS Compliant
The +Suffix identifies RoHS Compliance. See our web site for RoHS Compliance methodologies and qualifications

Maximum Ratings

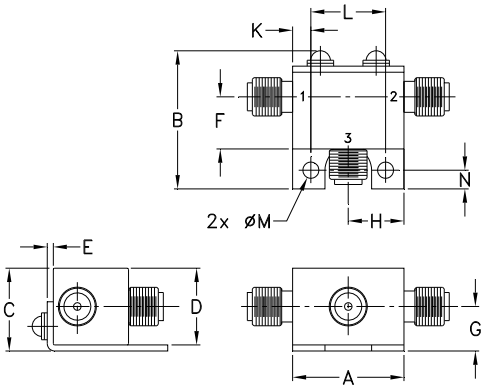
Operating Temperature	-40°C to 85°C
Storage Temperature	-55°C to 100°C
Power Input (as a splitter)	1.85W max.
Internal Dissipation	0.85 W max.
DC Current	0.4 A max.

Permanent damage may occur if any of these limits are exceeded.

Coaxial Connections

SUM PORT	S
PORT 1	1
PORT 2	2

Outline Drawing



Outline Dimensions (inch / mm)

A	B	C	D	E	F	G
.74	.90	.54	.50	.04	.34	.29
18.80	22.86	13.72	12.70	1.02	8.64	7.37

H	J	K	L	M	N	wt
.37	--	.122	.496	.106	.122	grams
9.40	--	3.10	12.60	2.69	3.10	20.0

Features

- wide bandwidh, 1800 to 12500 MHz
- excellent amplitude unbalance, 0.2 dB typ.
- good phase unbalance, 6 deg. typ.
- high ESD level
- DC passing
- protected under US patent 6,790,049

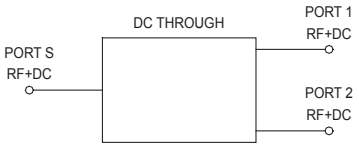
Applications

- WIMAX
- ISM
- instrumentation
- radar
- WLAN
- satellite communications
- LTE

Electrical Specifications at 25°C

Parameter	Frequency (MHz)	Min.	Typ.	Max.	Unit
Frequency		1800		12500	MHz
Insertion Loss (above theoretical 3.0 dB)	1800 - 3800	—	0.7	1.1	dB
	3800 - 8500	—	1.1	1.4	
	8500 - 12500	—	1.5	2.1	
Isolation	1800 - 3800	7.0	11.7	—	dB
	3800 - 8500	13.0	19.9	—	
	8500 - 12500	14.0	16.7	—	
Phase Unbalance	1800 - 3800	—	3.6	6.0	Degree
	3800 - 8500	—	8.1	12.0	
	8500 - 12500	—	11.5	—	
Amplitude Unbalance	1800 - 3800	—	0.1	0.2	dB
	3800 - 8500	—	0.1	0.4	
	8500 - 12500	—	0.3	0.9	
VSWR (Port S)	1800 - 3800	—	1.3	—	:1
	3800 - 8500	—	1.2	—	
	8500 - 12500	—	1.3	—	
VSWR (Port 1-2)	1800 - 3800	—	1.1	—	:1
	3800 - 8500	—	1.2	—	
	8500 - 12500	—	1.5	—	

Electrical Schematic



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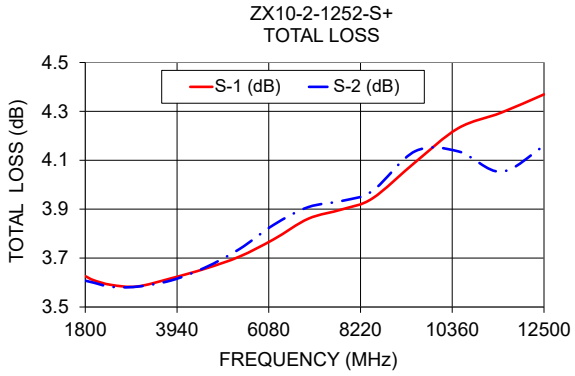
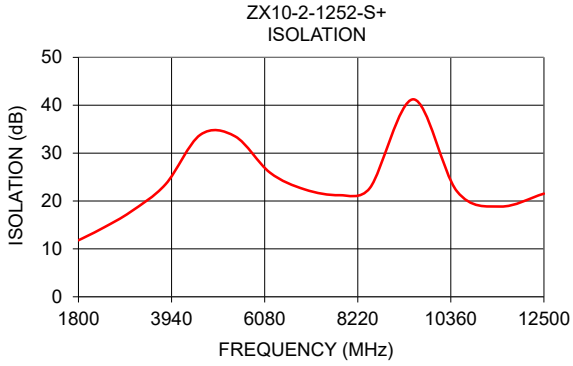
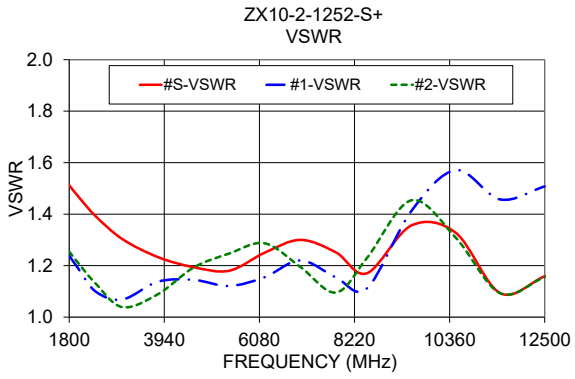
Typical Performance Data

Frequency (MHz)	Total Loss[1] (dB)		Amplitude Unbalance (dB)	Isolation (dB)	Phase Unbalance (deg.)	VSWR S	VSWR 1	VSWR 2
	S-1	S-2						
1800	3.63	3.61	0.02	11.80	0.92	1.51	1.24	1.25
2000	3.61	3.60	0.01	12.68	0.98	1.47	1.19	1.21
2400	3.59	3.58	0.01	14.55	1.16	1.39	1.10	1.13
3000	3.58	3.58	0.00	17.75	1.32	1.30	1.07	1.04
3800	3.62	3.61	0.01	23.48	1.74	1.23	1.14	1.09
4600	3.66	3.66	0.00	33.83	2.11	1.19	1.14	1.19
5400	3.71	3.74	0.03	33.51	2.52	1.18	1.12	1.25
6200	3.78	3.84	0.06	25.94	2.64	1.25	1.16	1.29
7000	3.86	3.91	0.05	22.42	2.70	1.30	1.22	1.20
7800	3.90	3.93	0.03	21.22	2.95	1.25	1.15	1.10
8500	3.94	3.97	0.03	22.72	3.42	1.17	1.11	1.23
9500	4.09	4.14	0.04	41.24	3.08	1.36	1.41	1.45
10500	4.23	4.14	0.09	22.04	3.37	1.33	1.57	1.31
11500	4.29	4.05	0.24	18.84	4.13	1.09	1.46	1.09
12500	4.37	4.16	0.21	21.54	5.17	1.16	1.51	1.16

1. Total Loss = Insertion Loss + 3dB splitter loss.







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